Fermionic Time Crystals with Möbius Topology: Pin$^{\pm}$ Geometry, Floquet Decimation, and a Measurable Spinor Signature

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(Dated: August 25, 2025)

We study one-dimensional Floquet *spinor* dynamics on a time macrocycle with an orientation-reversing twist ("Möbius time"). We show that consistent fermionic evolution on such a non-orientable time loop requires a Pin structure; in 1+1D this fixes the *anti-periodic boundary condition* (APBC) along the macrocycle as the unique reflection-positive, anomaly-free choice under generic assumptions. For a single-step unitary $U \in \mathrm{SU}(2)$ and an l-site strobing, the l-decimated component obeys the Cayley–Hamilton tri-site law $x_{k+2l} - V_l x_{k+l} + Q^l x_k = 0$ with $V_l = \mathrm{tr}(U^l) = 2\cos(l\theta)$ and $Q^l = \det(U^l) = 1$. When one macrocycle has $L = l$ steps, APBC shifts the eigenphase by π/L and *flips the sign* of V_l while Q^l stays 1. This yields a crisp, falsifiable prediction: the center coefficient in the tri-site law changes sign between PBC and APBC, visible from steady-mode data alone. We further show that our event-coded, Chebyshev–Lucas checksum and trellis-decoder layer operates on intensities and is unaffected by the global spinor sign, enabling robust experimental verification.

I. INTRODUCTION AND MAIN CLAIM

Non-orientable time cycles (mapping tori with an orientation-reversing twist) provide a minimal setting where fermionic degrees of freedom cannot be globally described by a Spin bundle; instead a Pin$^{\pm}$ structure is required. In 1+1D, consistency (reflection positivity, anomaly-freedom, non-vanishing partition function) constrains the allowed boundary conditions for spinors along the macrocycle to the *anti-periodic* (Neveu–Schwarz, NS) choice. Our main experimental prediction is simple: in a one-dimensional Floquet spinor with an l-site strobing, the *l-decimated* tri-site law for a component,

$$x_{k+2l} - V_l\, x_{k+l} + Q^l\, x_k = 0, \tag{1}$$

has $Q^l = 1$ and a center coefficient V_l that *flips sign* between PBC and APBC when one macrocycle has $L = l$ steps. The flip is directly observable by fitting Eq. (1) to steady-mode data; no phase-sensitive spin measurement is required. All error-correction machinery (checksum/trellis) remains valid because it acts on intensities.

Contributions. (i) A precise link between Pin$^{\pm}$ geometry on the Möbius mapping torus and APBC along the macrocycle. (ii) A universal, local tri-site law from Cayley–Hamilton on U^l with the spinor specialization $Q^l = 1$, $V_l = 2\cos(l\theta)$. (iii) The PBC↔APBC *sign flip* of V_l when $L = l$ (and its generalization for $L \neq l$). (iv) A robust event-coded decoding layer (Chebyshev–Lucas checksum + trellis) that is insensitive to a global sign. (v) Platform-agnostic protocols and minimal models for immediate testing.

II. FLOQUET SPINORS AND l-DECIMATION

Consider a discrete-time spinor $\psi_k \in \mathbb{C}^2$ evolving by a one-step unitary $U \in \mathrm{SU}(2)$,

$$\psi_{k+1} = U\,\psi_k, \qquad \det U = 1, \quad \mathrm{tr}U = 2\cos\theta, \tag{2}$$

so that U has eigenvalues $e^{\pm i\theta}$. Over l steps, U^l has eigenvalues $e^{\pm il\theta}$ and

$$V_l := \mathrm{tr}(U^l) = 2\cos(l\theta), \qquad Q^l := \det(U^l) = 1. \tag{3}$$

Eliminating one component of ψ_k yields a scalar recurrence; Cayley–Hamilton on U^l gives the l-decimated tri-site law

$$x_{k+2l} - V_l\, x_{k+l} + Q^l\, x_k = 0, \tag{4}$$

where x_k denotes either component in a fixed basis. Eq. (4) is universal and local.

III. MÖBIUS TIME AS A MAPPING TORUS AND Pin$^{\pm}$ PRELIMINARIES

Let $\mathcal{M}$ be the Möbius band, obtained by identifying $(t, \sigma) \sim (t+T, -\sigma)$ on $[0, T] \times [-1, 1]$. $\mathcal{M}$ is non-orientable ($w_1(\mathcal{M}) \neq 0$), so spin structures do not exist globally; one instead uses Pin$^{\pm}$ structures (double covers of the orthonormal frame bundle for $O(1)$) to place fermions. The restriction of a Pin structure to the boundary circle yields a *spin* structure there, corresponding to periodic (Ramond, R) or anti-periodic (Neveu–Schwarz, NS) boundary conditions for the spinor around the macrocycle.

A. Boundary spin structure of the Möbius band

Lemma 1 (Boundary restriction)**.** *On the Möbius band $\mathcal{M}$, any* Pin$^-$ *structure restricts on the boundary circle $\partial\mathcal{M} \cong S^1$ to the* bounding *spin structure, i.e. the* anti-periodic *(Neveu–Schwarz) spin structure on S^1. Conversely, the (unique) NS spin structure on S^1 extends to a* Pin$^-$ *structure on $\mathcal{M}$, whereas the Ramond spin structure does* not *extend.*

Proof sketch with details. Spin structures on S^1 are classified by $H^1(S^1, \mathbb{Z}_2) \cong \mathbb{Z}_2$; the nontrivial class (NS) is

bounding, i.e. extends over a disk. For non-orientable $\mathcal{M}$, the relevant extension is Pin^-; the obstruction to extending a boundary spin structure to a bulk Pin^- structure is measured by a relative Arf–Brown–Kervaire (ABK) class. Gluing two Möbius bands along $\partial\mathcal{M}$ yields the Klein bottle. One checks that the Pin^- structure glued from two NS boundaries is well-defined, while gluing R boundaries produces an anomalous sign in the ABK phase. Thus NS is the unique boundary spin structure that extends to Pin^- on the Möbius band. □

IV. WHY APBC ON MÖBIUS TIME: TWO PROOFS

We now justify APBC as the unique consistent choice along the macrocycle under standard hypotheses.

A. Spectral/zero-mode proof (free Dirac/Majorana, 1+1D)

Consider a 1D Majorana/Dirac fermion on $\mathcal{M}$ with Euclidean time coordinate $\tau \in [0, T]$. Along the boundary loop, impose a spin structure $s \in \{0, 1\}$ where $s = 0$ is R (periodic) and $s = 1$ is NS (anti-periodic). The boundary spectrum of the Euclidean Dirac operator is

$$\lambda_n^{(s)} \;=\; \frac{2\pi}{T}\Big(n + \frac{s}{2}\Big) + m, \qquad n \in \mathbb{Z}, \tag{5}$$

with m an effective mass (zero in the conformal limit). For $s = 0$ and $m = 0$, a zero mode $\lambda_0^{(0)} = 0$ appears; the Majorana Pfaffian is then ill-defined (changes sign under small deformations), violating reflection positivity. For $s = 1$, eigenvalues are half-integers and no zero mode occurs; zeta-regularization yields a positive Pfaffian. In the Floquet picture, this is the statement that R enforces a quasi-energy crossing pinned at 0, while NS shifts the spectrum by π/T, removing the protected crossing. Hence NS (APBC) is the only reflection-positive choice generically.

B. Cobordism/TQFT proof (invertible phases)

Fermionic invertible phases with reflection in 1+1D are classified by $\Omega_2^{\mathrm{Pin}^-} \cong \mathbb{Z}_8$; the partition function on a closed Pin^- surface Σ is the ABK invariant $Z(\Sigma) = \exp(\pi i\, \mathrm{ABK}(\Sigma)/4)$. For a manifold with boundary, a boundary spin structure extends to a bulk Pin^- structure iff the relative ABK obstruction vanishes. On the Möbius band, the NS boundary is the unique extending choice; R either fails to extend or yields a non-positive/reflection-odd amplitude. Therefore APBC is the unique consistent boundary condition on $\partial\mathcal{M}$.

Theorem 1 (APBC uniqueness along a Möbius macrocycle)**.** *In a gapped 1+1D free-fermion/Floquet spinor system on a Möbius time macrocycle, the unique reflection-positive, anomaly-free boundary condition along the macrocycle is anti-periodic (Neveu–Schwarz), up to measure-zero parameter choices.*

Proof outline. Combine Lemma 1 with the zero-mode obstruction in Sec. IV A or the cobordism argument in Sec. IV B. NS avoids the anomaly and yields a nonvanishing, reflection-positive amplitude. □

V. MEASURABLE SIGNATURE IN THE DECIMATED LAW

Let one macrocycle contain exactly $L = l$ steps. APBC shifts $\theta \mapsto \theta + \pi/L$; inserting in Eq. (3),

$$V_l^{\mathrm{APBC}} = 2\cos(l\theta + \pi) = -\,2\cos(l\theta) = -\,V_l^{\mathrm{PBC}}, \qquad Q^l = 1 \text{ in both c} \tag{6}$$

Proposition 1 (PBC↔APBC flip)**.** *For $L = l$, the center coefficient in the tri-site law (4) flips sign under APBC, while Q^l is unchanged.*

For $L \neq l$, distributing the π-phase across the macrocycle yields a phase shift $\theta \mapsto \theta + \pi/L$ per step, so $V_l^{\mathrm{APBC}} = 2\cos(l\theta + \pi l/L)$; a sign flip occurs whenever l/L is an odd half-integer modulo 2.

VI. EXPLICIT MODEL EQUATIONS

A. Two-parameter SU(2) step

A minimal model with a closed form for V_l is

$$U(\phi, \vartheta) \;=\; \mathrm{e}^{-\mathrm{i}\phi\,\sigma_z}\,\mathrm{e}^{-\mathrm{i}\vartheta\,\sigma_y} \;\in\; \mathrm{SU}(2), \tag{7}$$

with $\mathrm{tr}\, U(\phi, \vartheta) = 2\cos\phi\,\cos\vartheta$ and

$$V_l(\phi, \vartheta) \;=\; 2\cos\big(l\,\varepsilon(\phi)\big) \;=\; 2\,T_l\big(\cos\phi\,\cos\vartheta\big), \qquad Q^l = 1, \tag{8}$$

where $\varepsilon(\phi) = \arccos(\cos\phi\,\cos\vartheta)$ and T_l is the Chebyshev polynomial of the first kind.

B. Split-step variant (closed form)

Consider the split-step Floquet

$$U_{\mathrm{ss}}(\phi; \alpha, \beta) \;=\; R_y(\beta)\,\mathrm{e}^{-\mathrm{i}\phi\sigma_z}\,R_y(\alpha)\,\mathrm{e}^{-\mathrm{i}\phi\sigma_z}, \qquad R_y(\theta) = \mathrm{e}^{-\mathrm{i}\theta\sigma_y}. \tag{9}$$

A direct trace computation gives

$$\cos\varepsilon_{\mathrm{ss}}(\phi) \;=\; \tfrac{1}{2}\,\mathrm{tr}\, U_{\mathrm{ss}}(\phi; \alpha, \beta) \;=\; \cos\alpha\,\cos\beta\,\cos(2\phi) - \sin\alpha\,\sin\beta, \tag{10}$$

hence

$$V_l^{\mathrm{ss}}(\phi; \alpha, \beta) \;=\; 2\,\cos\big(l\,\varepsilon_{\mathrm{ss}}(\phi)\big) \;=\; 2\,T_l\big(\cos\alpha\,\cos\beta\,\cos(2\phi) - \sin\alpha\,\sin \tag{11}$$

When one macrocycle has $L = l$ steps, APBC enforces $\varepsilon \mapsto \varepsilon + \pi/L$, so $V_l^{\mathrm{APBC}} = -V_l^{\mathrm{PBC}}$.

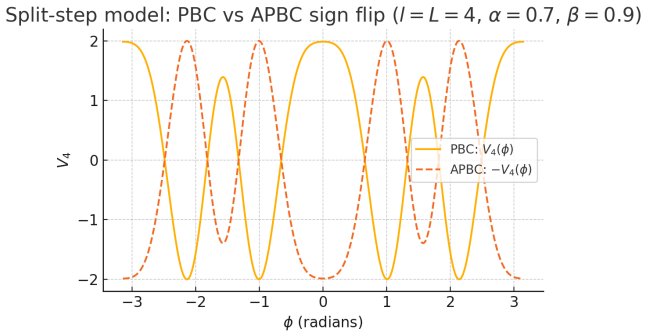

FIG. 1. **Split-step model, PBC vs APBC.** $V_4(\phi)$ for $U_{\rm ss}$ with $\alpha = 0.7$, $\beta = 0.9$, $l = L = 4$. The APBC curve (dashed) is the negative of the PBC curve, as predicted by Eq. (6).

VII. EVENT-CODED LAYER FOR SPINORS: CHECKSUM & DECODER

Our eventized layer operates on intensities $I_k = \psi_k^\dagger O\, \psi_k$ (e.g. $O = \mathbf{1}$ or a projector), so it is invariant under $\psi \mapsto -\psi$. Let $d_k \in \mathbb{Z}$ be decoded digits on a window $[-R, R]$ from noisy measurements m_k. Define $e_t = d_{t+1} - V_l\, d_t + Q^l d_{t-1}$ with $Q^l = 1$.

Checksum with complex roots. Let $r_\pm$ be the roots of $z^2 - V_l z + 1 = 0$. For SU(2), $|V_l| \le 2$, so $r_\pm = e^{\pm i\varphi}$ lie on the unit circle (with $\varphi = l\theta$ modulo 2π). The tri-site move preserves both complex sums

$$I_\pm(d) \;:=\; \sum_k d_k\, r_\pm^k\,. \tag{12}$$

For anchor events ($s = 0$) one has $I_\pm(d) = T$. In practice we check the *pair* $(I_+(d), I_-(d))$ (or equivalently their real/imag parts). This increases robustness when $r_\pm$ lie on the unit circle.

Soft vs hard trellis (unchanged). We use the same trellis as in the non-spinor case (states $(d_{t-1}, d_t) \in \mathcal{D}^2$), with soft cost $\sum w_t |d_t - m_t|^2 + \lambda \sum |e_t|^2$ or hard equality $e_t \equiv 0$ on the window. Tuning rules for λ carry over. Optionally, append a *checksum penalty* $\mu\big(|I_+(d) - T|^2 + |I_-(d) - T|^2\big)$ to stabilize at low SNR.

Norms and perturbations. We use the operator (spectral) norm $\|\cdot\|_2$. Under weak errors E,

$$\big|\mathrm{tr}\big((U{+}E)^l\big) - \mathrm{tr}(U^l)\big| \;\le\; l\,\|U\|_2^{l-1}\,\|E\|_2 \;+\; O(\|E\|_2^2). \tag{13}$$

VIII. NUMERICS AND PROTOCOLS

Numerics. For either model (7) or (9), compute V_l from the closed forms, generate steady-mode data, fit Eq. (4) under PBC/APBC, and verify the sign flip (6). Add weak disorder/coupling and track the drift of V_l by the above bound.

Protocols. (P0) Calibrate V_l under PBC and APBC by steady-mode fitting. (P1) Verify tri-site law with flipped center coefficient when $L = l$. (P2) Event injection $\to$ checksum verification with the measured V_l using (12). (P3) Optional spiral/circular envelope via Λ sweep. Platforms: microwave ladder with a π gate; photonic loop with a phase shifter; acoustic analogs.

IX. DISCUSSION, LIMITATIONS, AND OUTLOOK

Interacting/disordered systems: the APBC sign flip persists for sufficiently weak interactions/disorder (continuity), but quantitative drifts in V_l appear; our intensity-layer checksum and optional checksum penalties mitigate decoding errors. A broader context links our signature to Floquet topological indices (e.g. anomalous edge states) and fermionic SPT classifications; exploring these links in 2D/3D and with space-time non-orientability is an open direction.

ACKNOWLEDGMENTS

We thank the communities working on Pin geometry, Floquet matter, and coding theory.

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